REAL-WORLD ASSETS (RWA)

Tokenizing *real* assets, on-chain.

How treasuries, credit, equities, and real estate are being moved onto programmable rails.

A *real-world asset*, represented on-chain.

The industry term for any off-chain asset wrapped into a token.

WHAT IT IS

A token on a blockchain that *legally represents* an off-chain asset.

The token is the on-chain wrapper. The asset still lives off-chain — held in custody, registered in a legal entity, governed by securities law.

WHY IT MATTERS

24/7 settlement, fractional ownership, composable with stablecoins.

The wrapper turns illiquid, batch-settled instruments into 24/7 programmable money. Treasuries, bonds, funds, real estate become instantly transferable collateral.

THE SHAPE OF IT



Five layers between the *asset* and the *wallet*.

A real-world asset (RWA) is wrapped into a token. Five layers stand between the off-chain claim and the investor's wallet.

LAYER 0

The *real-world asset*

The off-chain claim being represented on-chain — a bond, fund unit, property deed, equity stake or commodity. Exists in the legal world before any token does.

LAYER 1 · LEGAL WRAPPER

SPV / Trust / Fund

A regulated entity owns the real-world asset and issues claims against it. Defines investor eligibility and bankruptcy remoteness.

LAYER 2 · CUSTODY

Off-chain custodian

Holds the underlying — Treasuries at BNY Mellon, USD at a money-center bank, deeds at a title company. Audited monthly.

LAYER 3 · SMART CONTRACT / ISSUANCE PLATFORM

The token, with rules

ERC-20 / ERC-1400 / ERC-3643 contract enforcing transfer eligibility, KYC gating, redemption and corporate actions. Minted via Securitize, Tokeny or Provenance.

LAYER 4 · BLOCKCHAIN

Settlement layer

Public (Ethereum, Solana) or permissioned (Canton, Onyx, Fnality). Provides ordering, finality and an open API.

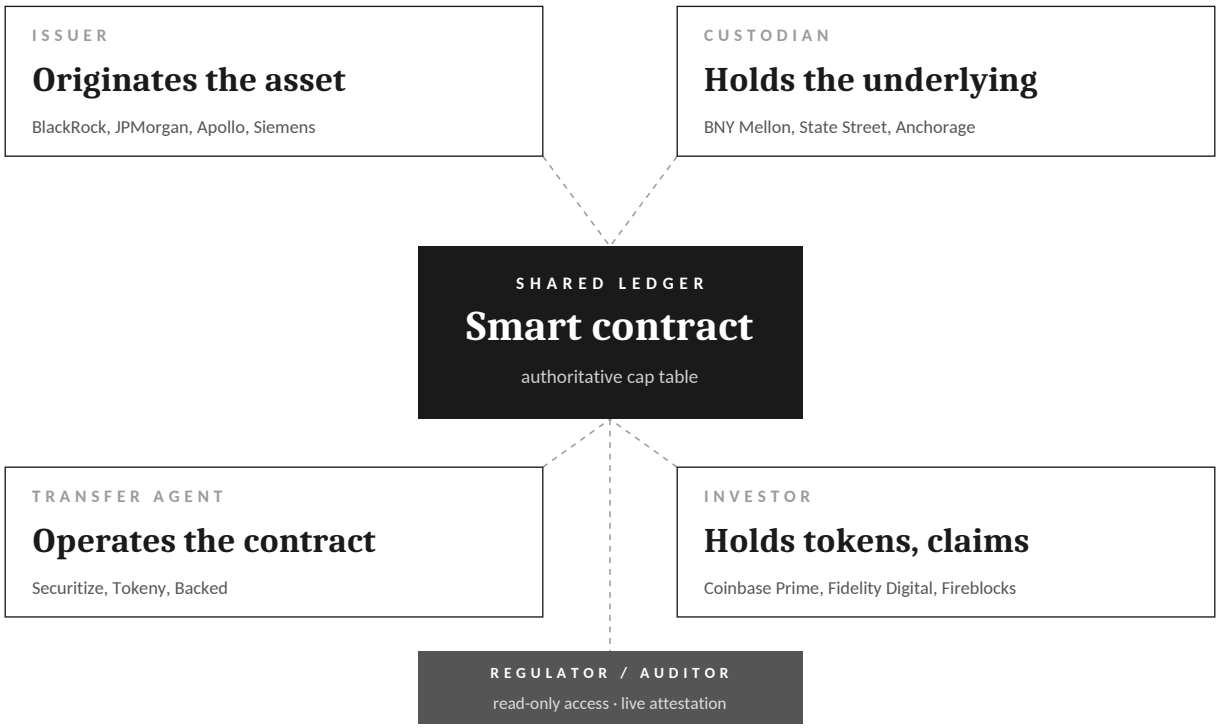
LAYER 5 · WALLET / DISTRIBUTION

Investor endpoint

Institutional custody (Fireblocks, Anchorage), broker app, or self-custody wallet — distributed via Coinbase Prime, Securitize Markets, regulated venues.

The same actors as today. A *different rendezvous*.

Tokenization does not eliminate the parties involved in a securities issuance — it changes where they meet. Instead of reconciling private databases via SWIFT messages and end-of-day batches, each party reads from and writes to the same ledger. The smart contract becomes the authoritative cap table.

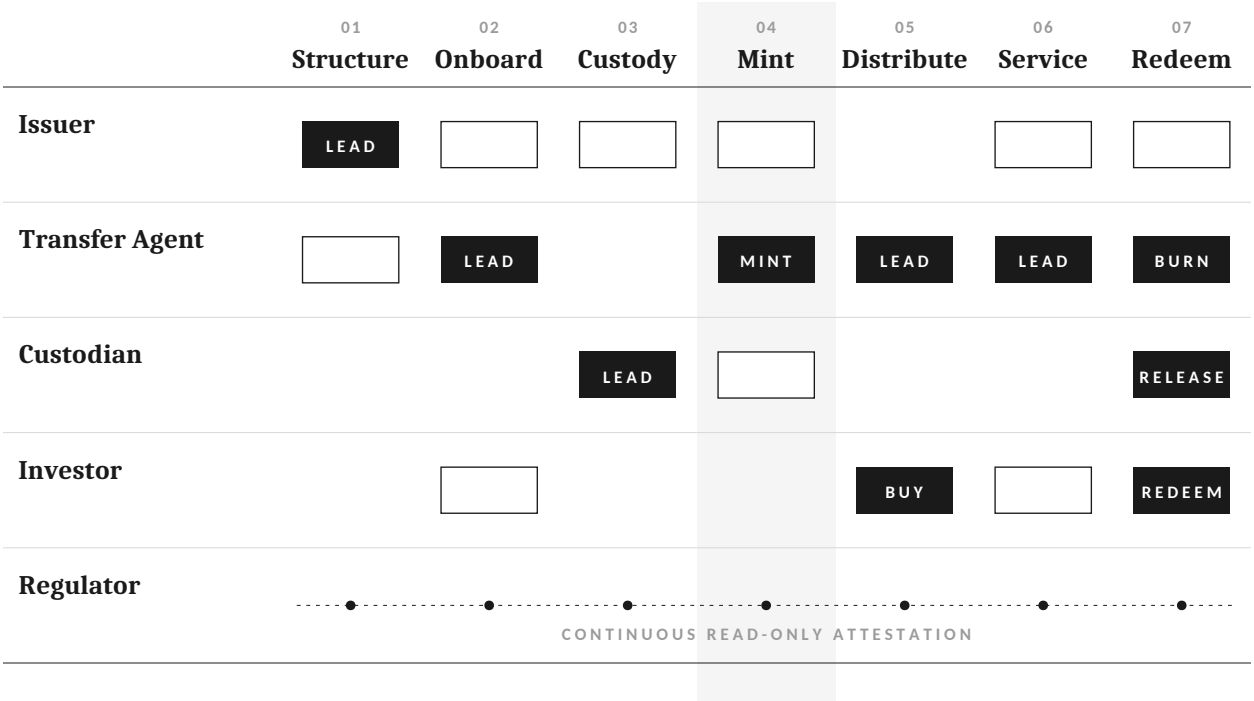


THE FUNCTIONAL SHIFT

Today, each spoke maintains its own copy of the truth and reconciles overnight. On a shared ledger, there is one copy and reconciliation is implicit. The savings are operational, not philosophical.

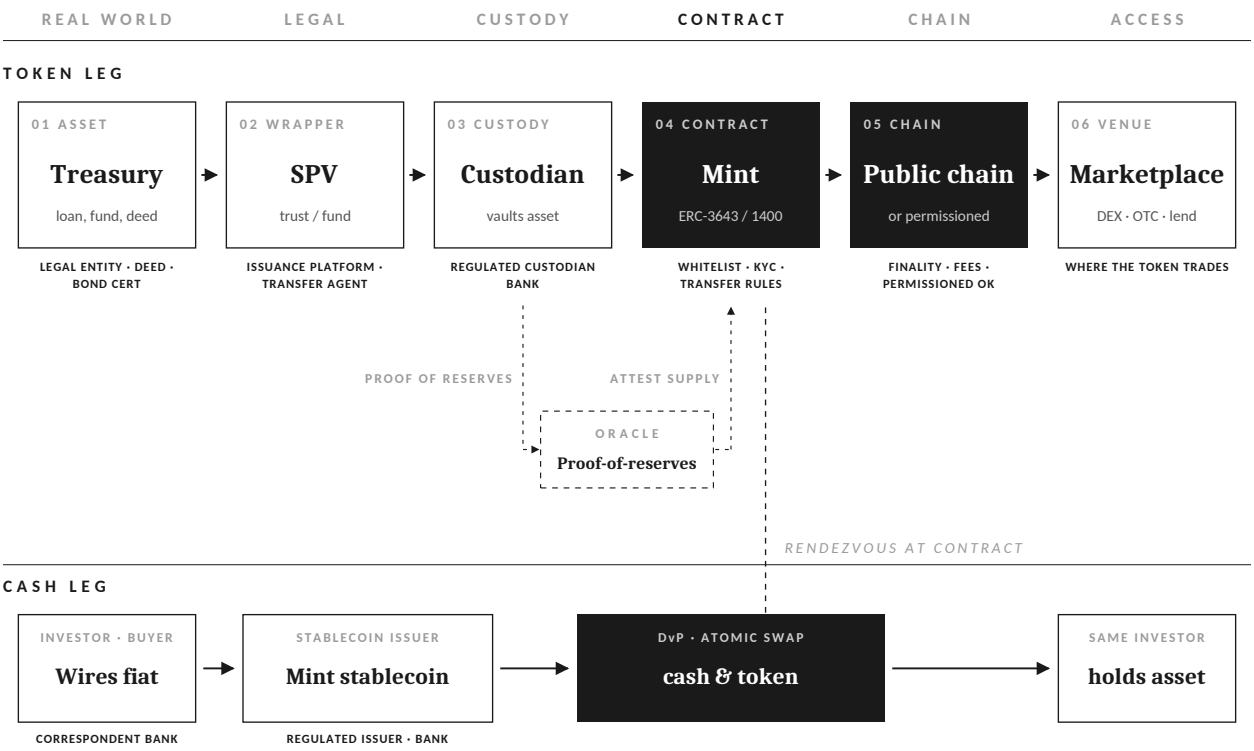
Seven steps. *Five parties.* One activity graph.

A walkthrough of an end-to-end issuance. The novel step is step 04 (Mint). Everything else is paperwork, custody, and disclosure that already exists in conventional finance.



A bond enters from *the left*. A token clears on *the right*.

Synthesis of every layer in one diagram — legal, custody, oracle, contract, settlement, access — with the parallel cash leg shown beneath.



TOKEN MOVEMENT VALUE-CRITICAL ARROW ATTESTATION / PROOF

04 CONTRACT IS THE NOVEL STEP. EVERYTHING ELSE IS PAPERWORK YOU ALREADY DO.

Six standards do almost *all the work.*

A “token standard” is a published interface — a contract template every wallet and custodian already knows how to read. For tokenized real-world assets, the choice is largely between a plain fungible token and one with compliance baked in.

STANDARD	FAMILY	WHAT IT ENFORCES	WHERE YOU SEE IT
ERC-20	Fungible	Identical, divisible units. No transfer rules. The plain workhorse.	Stablecoins · tokenized money-market funds · most yield wrappers
ERC-721	Non-fungible	Each token is unique with its own ID and metadata.	Real-estate deeds · individual loans · art · collectibles
ERC-1155	Multi-token	Many fungible and non-fungible series in one contract. Cheap batch transfers.	Pools of receivables · gaming-style asset packs · invoice batches
ERC-3643	Permissioned	On-chain identity registry. Every transfer checks KYC, accreditation, jurisdiction. The de-facto institutional security-token standard.	Tokenized equities · regulated funds · private credit · sukuk
ERC-1400	Permissioned	Earlier security-token framework. Partitions (e.g. share classes) and forced transfers for regulators. Still in production but losing share to	Legacy security-token issuance platforms
ERC-4626	Vault wrapper	Standard interface for yield-bearing vaults — deposit, withdraw, share-price math.	Tokenized funds, structured-yield strategies, restaking products

PICK BY ASSET CLASS

Treasuries · MMFs
ERC-20 + 4626

Tokenized equity
ERC-3643

Private credit
ERC-3643 / 1400

Real estate deed
ERC-721

Loan pool
ERC-1155

Stablecoin (cash leg)
ERC-20

Beyond Ethereum. Solana's Token-2022 bakes transfer-hooks and confidential balances into the base token; Stellar uses SEP-8 regulated assets; Algorand ships compliance flags natively. Same pattern — fungible base plus a permissioning layer for anything regulated.

Six buckets that count as RWA. *Plus* the cash leg.

"RWA" by industry convention excludes stablecoins — those are the on-chain cash leg, the settlement asset against which everything below trades. Within RWA proper, tokenized money-market funds and private credit dominate volume. Real estate, commodities, equities, and collectibles are smaller but where the structural arguments for tokenization are most interesting.

RWA TOTAL · ~$28B

01 · LARGEST

Private credit.

$13B

~46% OF RWA

Loans · receivables · consumer credit

MAPLE · CENTRIFUGE · GOLDFINCH

02 · INSTITUTIONAL

Treasuries / MMF.

$11B

~39% OF RWA

US T-bills · money-market funds

BUIDL · BENJI · OUSG · USDY

03

Equities

$2.0B

BACKED · DINARI · INTAIN

04

Commodities

$1.5B

PAXG · XAUT · CARBON

05

Real estate

$0.4B

PROPCHAIN · LOFTY

06

Collectibles

$0.2B

MASTERWORKS

— NOT RWA · SETTLEMENT LEG

Stablecoins **$240B** — the on-chain cash against which RWA trades clear. USDC · USDT · PYUSD

Top tier (private credit + tokenized treasuries) is ~85% of all RWA on-chain today. Equities and real estate are the two segments where every projection adds the most upside — and where the regulatory and legal-wrapper work is hardest.

Source: rwa.xyz, DefiLlama, primary issuer disclosures.

Three camps, increasingly *overlapping*.

CAMP 1 Incumbent issuers

Real assets, regulated, deep capital

BlackRock

BUIDL — tokenized money-market fund on Ethereum, Aptos, Avalanche. Largest by AUM.

$2.9B AUM

Franklin Templeton

BENJI / FOBXX — first US-registered '40 Act fund on a public chain.

$650M AUM

JPMorgan

Onyx / Kinexys — tokenized collateral, intraday repo, FX PvP for institutions.

$1.5T CUMULATIVE

Goldman Sachs

GS DAP — tokenized issuance and trading platform for tokenized debt; EIB digital bond.

PERMISSIONED

CAMP 2 Crypto-native issuers

On-chain first, building toward institutional

Ondo Finance

OUSG, USDY — tokenized Treasuries wrapping BlackRock and Franklin funds.

$1.4B AUM

Maple Finance

Institutional credit pools — secured lending, T-Bill products.

$2.1B AUM

Centrifuge

Tokenized private credit and structured products with on-chain waterfalls.

$650M AUM

Backed

Tokenized public equities and ETFs, EU-issued, EU-regulated.

EU MiCA

CAMP 3 Infrastructure

Issuance, transfer-agency, custody, chains

Securitize

Broker-dealer + transfer agent. The plumbing under BUIDL.

SEC-REGISTERED

Fireblocks

Institutional custody and tokenization tooling — used by major banks.

CUSTODY

Canton Network

Privacy-preserving permissioned chain — favored by global banks.

CHAIN

Chainlink

Cross-chain messaging, NAV oracles, proof-of-reserves for tokenized assets.

MIDDLEWARE

The token is novel. The *claim* is not.

The dominant regulatory thesis: a tokenized security is still a security; a tokenized fund is still a fund. Existing rules apply, with bolted-on requirements for custody, recordkeeping, and crypto-asset disclosure. Where regulators differ is on speed, on whether public chains are eligible rails, and on stablecoin treatment.

EU	**MiCA** in force since Dec 2024 covers crypto-assets (incl. asset-referenced and e-money tokens). The DLT Pilot Regime allows MTFs and CSDs to settle tokenized	PERMISSIVE
United States	**FIT21 + GENIUS Act** (2025) clarified stablecoin and market-structure rules. SEC has approved tokenized fund products; CFTC oversees commodity-tokens. Fragmented	EVOLVING
United Kingdom	**FCA Digital Securities Sandbox** live since 2024. Stablecoin regime in 2025. Fund-tokenization roadmap published by HM Treasury / Investment Association.	PERMISSIVE
Singapore	**MAS Project Guardian** the most-cited industrial-scale pilot — bond, FX, and asset-management tokenization with global banks. Production licences for stablecoins	PERMISSIVE
UAE	Abu Dhabi (FSRA) and Dubai (VARA) operate parallel regimes. Aggressive issuer onboarding, including for real-estate and private-credit tokens.	PERMISSIVE
Hong Kong	**SFC** circular permits tokenized funds; HKMA Project Ensemble tests wholesale CBDC-settled tokenized assets. Retail stablecoin licensing in force.	PERMISSIVE
Switzerland	**DLT Act** (2021) established ledger-based securities at law. SDX operates a regulated tokenized exchange. Quietly the most mature framework.	PERMISSIVE
Japan	FIEA amendments treat security tokens as Type-1 financial instruments. Real-estate and corporate-bond issuance active; stablecoin framework live since 2023.	EVOLVING
China	Mainland prohibits crypto. mBridge participation (CBDC) suspended 2024. HK is the de-facto onshore venue.	RESTRICTIVE

Risks & *criticisms*.

It is fair to read the section above and ask what is actually new here. Several thoughtful objections deserve to be taken at face value.

A solution looking for a problem.

Many tokenized assets simply migrate an existing intermediated structure onto a different ledger. Without a true on-chain investor base, the savings are theoretical.

Liquidity does not appear by decree.

Tokenizing an illiquid asset gives you a tokenized illiquid asset. Secondary markets remain thin outside of stablecoins and tokenized Treasuries.

Smart-contract and oracle risk.

Code defects, upgrade-key compromise, and price-feed manipulation introduce failure modes that traditional infrastructure does not have.

Off-chain enforcement gap.

The chain says you own a building; a court still settles the dispute. Legal-wrapper soundness is the entire trust assumption.

Fragmentation across chains.

Today's tokenized assets live on ten different settlement layers. Bridging adds risk and undermines the “single source of truth” argument.

Outlook to *2030*.

Three things to watch. None of them require believing crypto wins everything; all of them are plausible without it.

Cash legs go first.

Tokenized deposits and stablecoins displace correspondent-banking rails for B2B settlement. Highest-confidence prediction in the space.

Collateral mobility becomes the killer app.

Intraday-repo, cross-margin, and triparty collateral on a shared ledger save measurable basis points — the use case banks like JPM, Goldman, and HSBC are already in production on.

Funds before single-name equities.

Tokenized money-market and ETF wrappers will scale before tokenized Apple or Toyota, because the unit of account is already a fund share — not a dematerialized certificate.

Public chains earn institutional trust — slowly.

Permissioned chains (Canton, Onyx, Fnality) will dominate near-term volume. By the late 2020s, public-chain settlement of regulated assets becomes routine.

A boring conclusion is the bullish one.

The end-state is not a parallel financial system. It is the existing one, with shorter settlement cycles, fewer reconciliations, and a different vendor stack.

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Thinking about tokenizing your book? We're happy to dig in.

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Tokenization roadmap, asset selection, structuring of the wrapper.

ADVISORY

Choose the right issuance platform and chain. Negotiate terms.

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Banks, asset managers and issuers across EMEA and Asia.

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hello@lionscraft.io

lionscraft.io